

ECA15 TRANSMISSION LINE & WAVEGUIDES

EXP 6

Test Casse 1

```
clc;
```

```
clear;
```

```
close;
```

```
// Constants
```

```
sigma = 5.8e7;    // Conductivity of copper (S/m)
```

```
mu0 = 4 * %pi * 1e-7;
```

```
f101 = 3.335e9;   // Resonant frequency (Hz)
```

```
// Dimensions (meters)
```

```
a = input("Enter dimension a (m): ");
```

```
b = input("Enter dimension b (m): ");
```

```
c = input("Enter dimension c (m): ");
```

```
// Skin depth
```

```
d = sqrt(1 / (%pi * f101 * mu0 * sigma));
```

```
// Q-factor for TE101 mode
```

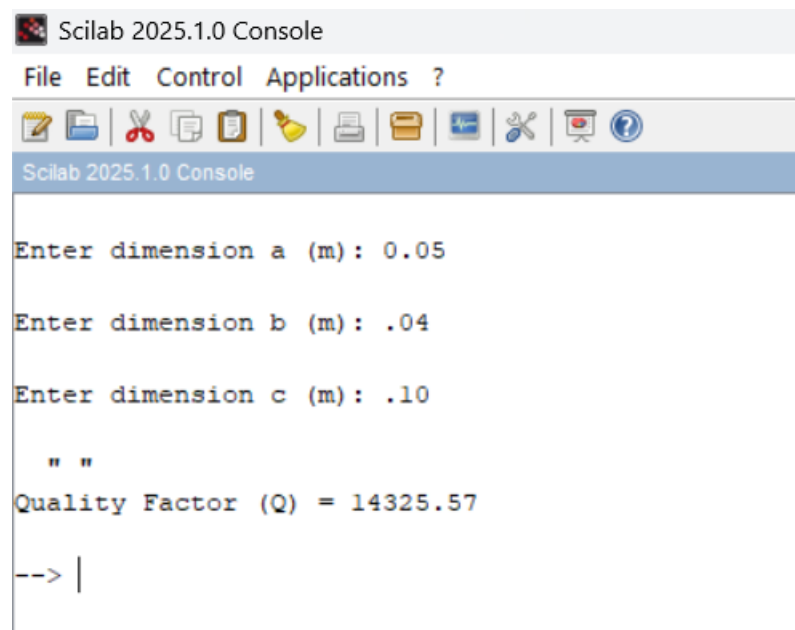
```
Q = ((a^2 + c^2) * a * b * c) / ...
```

```
(d * (2 * b * (a^3 + c^3) + a * c * (a^2 + c^2)));
```

```
disp(" ");
```

```
printf("Quality Factor (Q) = %.2f\n", Q);
```

Output:



```
Scilab 2025.1.0 Console
File Edit Control Applications ?
[Icons]
Scilab 2025.1.0 Console

Enter dimension a (m): 0.05

Enter dimension b (m): .04

Enter dimension c (m): .10

" "
Quality Factor (Q) = 14325.57

--> |
```

Test case 2:

```
clc;
```

```
clear;
```

```
sigma = 5.8e7;
```

```
mu0 = 4*%pi*1e-7;
```

```
f101 = 3.335e9;
```

```
a = 0.07;
```

```
b = 0.05;
```

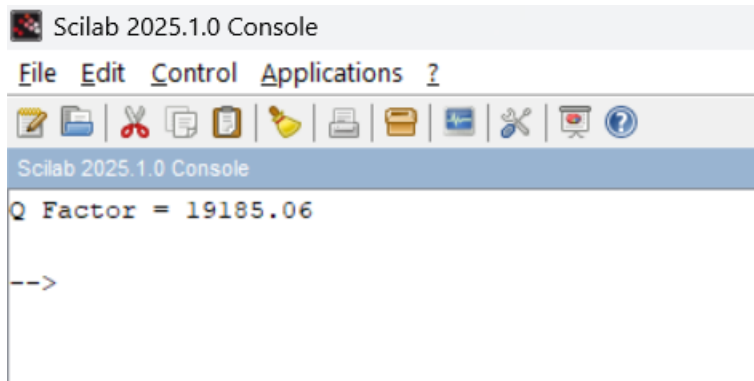
```
c = 0.12;
```

```
d = sqrt(1/(%pi*f101*mu0*sigma));
```

```
Q = ((a^2+c^2)*a*b*c)/(d*(2*b*(a^3+c^3)+a*c*(a^2+c^2)));
```

```
printf("Q Factor = %.2f\n",Q);
```

Output



The image shows a screenshot of the Scilab 2025.1.0 Console window. The window has a title bar that reads "Scilab 2025.1.0 Console". Below the title bar is a menu bar with the following items: "File", "Edit", "Control", "Applications", and "?". Below the menu bar is a toolbar with various icons for file operations (new, open, save, print, etc.) and editing (copy, paste, undo, redo, etc.). The main area of the console displays the text "Q Factor = 19185.06" on the first line and "-->" on the second line, indicating a prompt for further input.

```
Scilab 2025.1.0 Console
File Edit Control Applications ?
Q Factor = 19185.06
-->
```